EXPERIMENT 16:

AIM: To write a C program that performs a letter frequency attack on a monoalphabetic substitution cipher and displays the top 10 possible plaintexts in order of likelihood.

CODE:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int main() {
    char c[1000], p[1000];
    char freq[] = "ETAOINSHRDLCUMWFGYPBVKJXQZ";
    int f[26] = {0}, i, j, n, top;

    printf("Enter ciphertext: ");
    fgets(c, sizeof(c), stdin);

    printf("Enter number of top plaintexts: ");
    scanf("%d", &top);

    for (i = 0; c[i]; i++)
        if (isalpha(c[i]))
            f[toupper(c[i]) - 'A']++;
```

```
for (i = 0; i < 26; i++)
    for (j = i + 1; j < 26; j++)
        if (f[freq[i] - 'A'] < f[freq[j] - 'A']) {
            char t = freq[i];
            freq[i] = freq[j];
            freq[j] = t;
        }

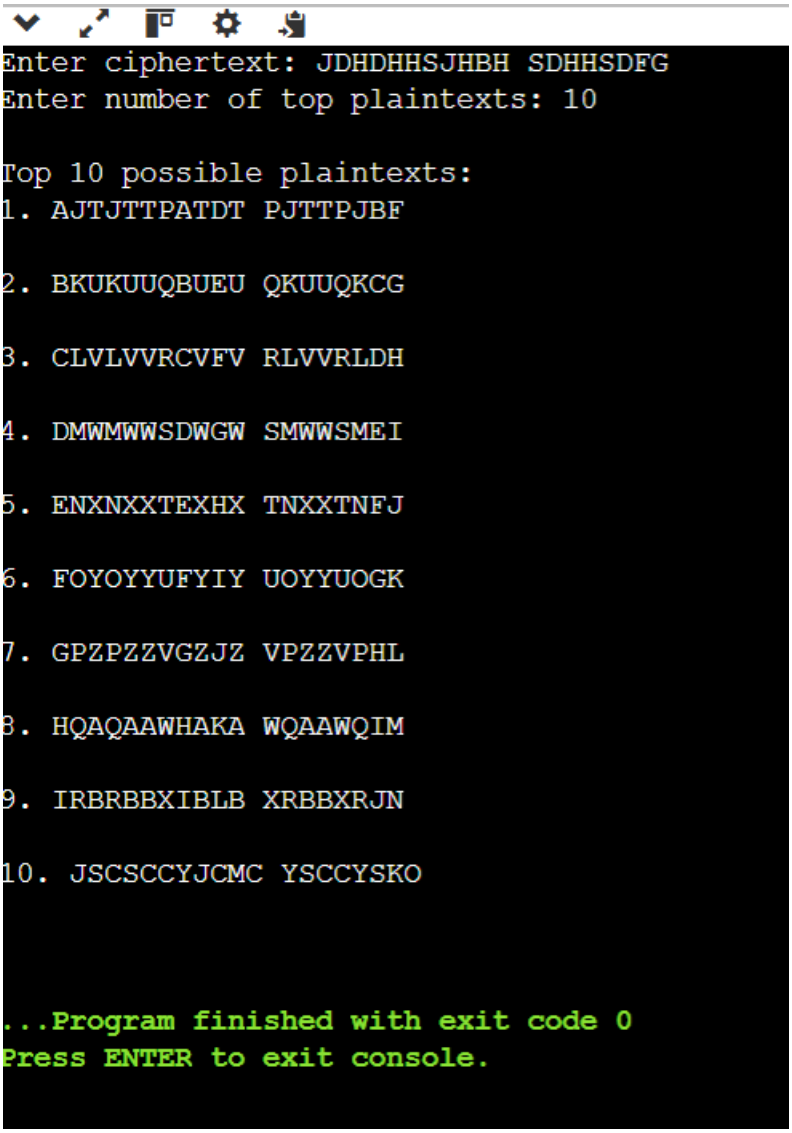
printf("\nTop %d possible plaintexts:\n", top);

for (n = 0; n < top && n < 10; n++) {
    for (i = 0; c[i]; i++) {
        if (isalpha(c[i])) {
            int x = toupper(c[i]) - 'A';
            p[i] = freq[x];
            if (islower(c[i]))
                p[i] = tolower(p[i]);
        } else
            p[i] = c[i];
    }
    p[i] = '\0';
```

```
        printf("%d. %s\n", n + 1, p);

        for (i = 0; i < 26; i++)
            freq[i] = (freq[i] - 'A' + 1) % 26 + 'A';
    }
    return 0;
}
```

OUTPUT:


Enter ciphertext: JDHDHHSJHBH SDHHSDFG
Enter number of top plaintexts: 10
Top 10 possible plaintexts:
1. AJTJTTPATDT PJTTPJBF
2. BKUKUUQBUEU QKUUQKCG
3. CLVLVVRCVFV RLVVRLDH
4. DMWMWWSDWGW SMWWSMEI
5. ENXNXXTEXHX TNXXTNFJ
6. FOYOYYUFYIY UOYYUOGK
7. GPZPZZVGZJZ VPZZVPHL
8. HQAQAAWHAKA WQAAWQIM
9. IRBRBBXIBLB XRBBXRJN
10. JSCSCCYJCMC YSCCYSKO
...Program finished with exit code 0
Press ENTER to exit console.